

WIREFLUID HACKATHON 2025

PSL INTEGRITY SHIELD

Transparent. Immutable.

Community-Governed Match Integrity.

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 github.com/mufeezhanif/wirefluid-hackathon (branch: zayan)



THE PROBLEM

Match-fixing is the single biggest threat to cricket's credibility in Pakistan.

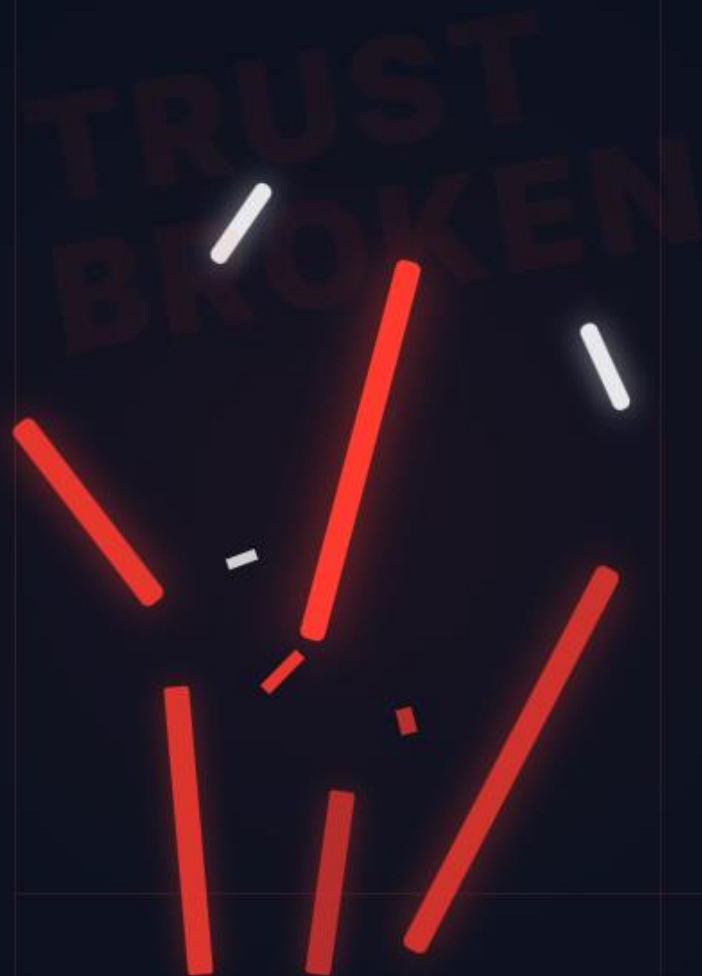
The 2010 spot-fixing scandal, repeated investigations, and ongoing suspicions have severely eroded fan trust. Current integrity systems are fundamentally flawed:

OPAQUE Investigations happen behind closed doors.

CENTRALIZED A single authority decides what's suspicious.

RETROACTIVE Problems are caught only after the damage is done.

ERASABLE Centralized records can be altered or suppressed.



OUR SOLUTION

PSL Integrity Shield: Three interlocking smart contracts that make match integrity transparent, immutable, and community-governed.

"If the blockchain can't be changed, neither can the truth about the match."



MatchOracle

Immutable event ledger.

Staked reporters submit ball-by-ball event hashes during live matches, creating a cryptographically secure, permanent record.



PredictionEngine

Commit-reveal mechanics.

Fans lock cryptographically hashed predictions before matches. Crowd intelligence is revealed only after the match, mathematically tamper-proof.

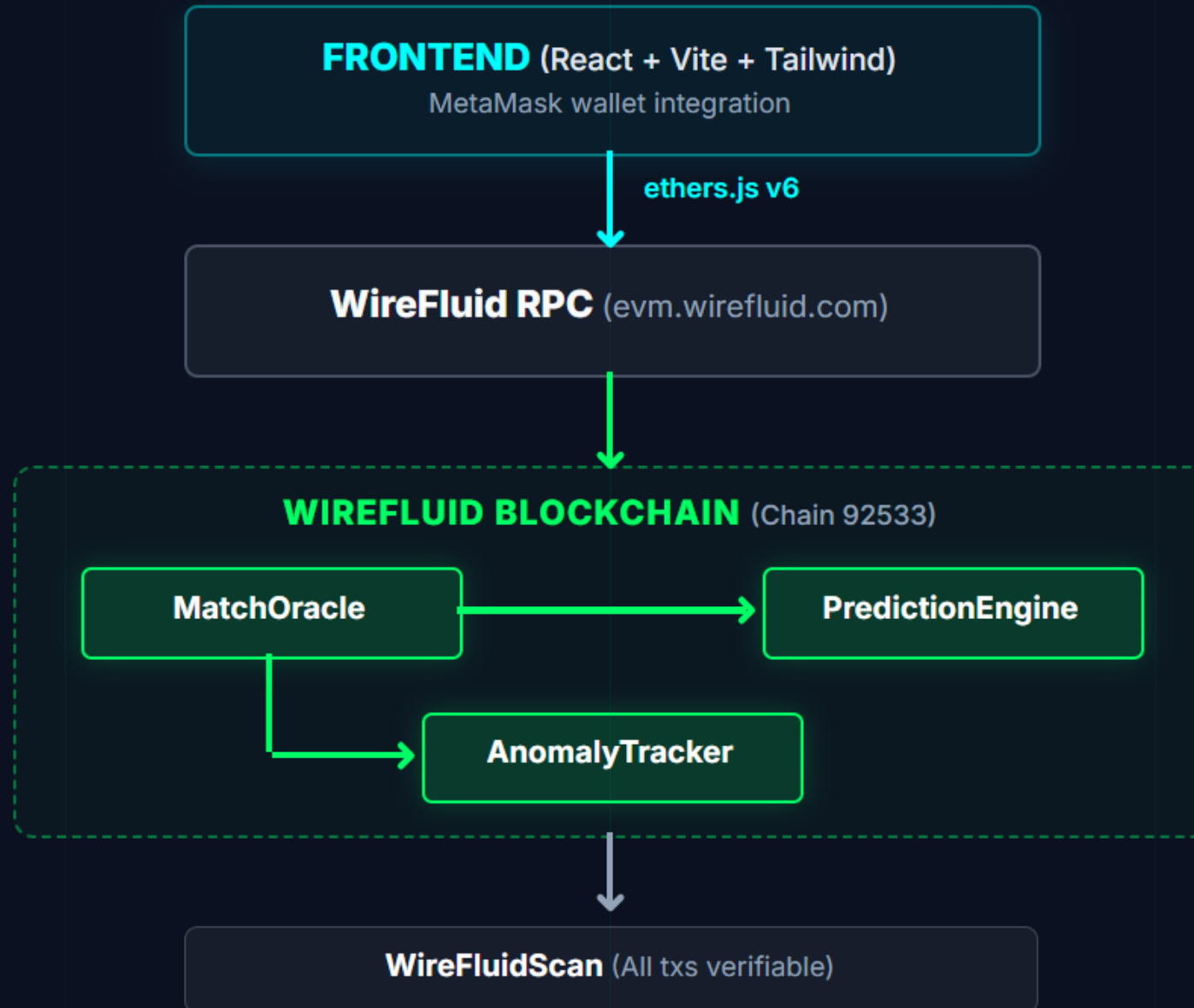


AnomalyTracker

Community governance.

Anyone can stake WIRE to flag suspicious matches. A decentralized, stake-weighted voting system resolves flags and updates integrity scores.

HOW IT WORKS (ARCHITECTURE)



WireFluid Network **Integration**

CRITICAL

Chain ID: 92533 **Currency:** WIRE

RPC URL: <https://evm.wirefluid.com>

Explorer: <https://scan.wirefluid.com>

Deployed Contracts

MatchOracle

`0x7b457E75Ad7AB2331FCDdeC5f0a80d33a6e34771`

PredictionEngine

`0xa97094e559e2570A37045542fed7A1Ff03763E87`

AnomalyTracker

`0x1570200C79D6245ba77797541e4fab90bCfDAC34`

Why WireFluid?

 **~\$0.01/tx costs** make real-time, ball-by-ball event recording economically viable on-chain.

 **~5 second finality** enables near-live match tracking without sacrificing user experience.

 **Transparent Explorer:** All protocol transactions are publicly verifiable on WireFluidScan.

\$ **Optimized Protocol Gas Costs:**

submitEvent:	~75K gas
revealPrediction:	~187K gas
voteOnFlag:	~105K gas

Key Features / Part 1: Core Protocol

1. Immutable Event Ledger

Staked reporters submit ball-by-ball event hashes during live matches. Events are **cryptographically fingerprinted** and timestamped on-chain. No one can alter records — not reporters, the league, or developers.

2. Commit-Reveal Predictions

Fans commit hashed predictions before matches using `keccak256(winner + runs + salt)`. Revealed only after match completion. Mathematically tamper-proof.

3. Community Anomaly Flagging

Anyone stakes WIRE to flag suspicious matches. Uses **stake-weighted voting** with a quorum of 3. Upheld flags return stake and drop integrity score; rejected flags result in slashed stakes (anti-spam).

4. Reporter Trust Scores

On-chain reputation ranging 0–200. Starts at 100, +1 per submitted event, -50 penalty on slashing. Color-coded trust badges visually rank reporters:

Green (≥ 80)

Yellow (50–79)

Red (< 50)

Key Features / Part 2: Analytics & Governance

\$ 5. Crowd Divergence

Auto-compares actual results vs crowd consensus. Alerts when divergence is **>70%** — signaling a highly suspicious outcome that warrants investigation.

✓ 6. Clean Match Cert

Matches with an integrity score **≥ 90** are certified on-chain. Emits a CleanMatchCertified event — a permanent proof of fairness.

📊 7. Season Dashboard

Aggregated analytics: matches tracked, average integrity score, total flags, and total predictions. Distributed clearly into Clean, Suspicious, or Flagged buckets.

★ 8. Accuracy Ranking

Ranks the top 10 predictors by accuracy percentage. Gamifies fan participation while generating valuable crowd-intelligence data for the protocol.

📄 9. Immutable Audit Trail

Every protocol event is permanently queryable: hash, reporter address, trust score, block number, and timestamp. Direct tx links to [WireFluidScan](#).

👤 10. Reporter Registry

A comprehensive public panel to view all staked reporters, check their active status, verify their color-coded trust scores, and monitor registration details.

Tech Stack / Testing & Security

Smart Contracts

Solidity 0.8.20 • OpenZeppelin v5.6.1
Hardhat 2.28.6

Frontend App

React 18.3.1 • Vite 6.0.5
ethers.js v6.9.0 • Tailwind 3.4.17

Blockchain Infrastructure

WireFluid Testnet Chain 92533
Wallet Integration — MetaMask (EIP-1193)

Testing & Validation

32/32
TESTS PASSING

MatchOracle	11 Tests
PredictionEngine	9 Tests
AnomalyTracker	12 Tests

Security Architecture

ReentrancyGuard: Applied to all state-changing and payable functions to prevent recursive exploit vectors.

Role-Based Access: Owner-only administrative rights via OpenZeppelin's robust access controls.

Require-Based State Validation: Strict lifecycle checks (e.g., predictions only before match start).

Economic Slashing & Quorums: Bad actors face strict stake slashing. Anti-griefing mechanisms enforce a strict quorum of 3 for governance actions.

Live Demo / Product Proof

 **PSL Shield**
ENTANGLED HACKATHON '26

DashboardSeason ReportPredictionsReportersLeaderboardAudit TrailWireFluid Testnet

LIVE ON TESTNET


Loading on-chain data...

 **ON-CHAIN CRICKET INTEGRITY**

No fixing. No cap.
Community-powered match integrity on **WireFluid Testnet**. Flag sus behaviour, stake WIRE, get rewarded. Frfr.

 **0**
Live Bn

 **0**
Total Flags

 **0**
Predictions

 **0**
Certified Clean

 **0**
Wrapped Up

All LiveUpcomingCompleted


No matches found

PROTOCOL IN ACTION

-  Live Match Dashboards
-  Commit-Reveal Flow
-  Accuracy Leaderboards
-  Season Integrity Analytics
-  Immutable Audit Trail
-  Reporter Registry

 **100% Verifiable via WireFluidScan**

Real-World Impact / Stakeholders



For the League (PCB/PSL)

Transparent integrity system that proves fairness to fans, sponsors, and broadcasters, restoring complete trust post-scandal.



For Reporters & Journalists

Economic incentives (stake + rewards) for accurate, real-time reporting while building an immutable, on-chain reputation they own.



For the Fans

Gamified participation through predictions and leaderboards, coupled with verification power and community-governed anomaly voting.



For Regulators (ICC / ACU)

Permanent, immutable on-chain records for investigations. No ambiguity, no erased data, and no "lost" match records.



Scalability Beyond PSL

Protocol is sport-agnostic. The core contracts can be instantly deployed to secure IPL, BBL, CPL, or any other professional league globally.

Roadmap / What's Next

01 Mainnet Deployment

Full production launch of all three smart contracts directly onto the **WireFluid Mainnet**.

02 Mobile Application

Dedicated mobile app with **push notifications** for live match events, anomalies, and prediction results.

03 AI Anomaly Detection

AI-powered detection layer feeding autonomous, data-driven flag proposals directly into the **AnomalyTracker**.

04 Multi-League Scaling

Global expansion scaling the protocol architecture to support **IPL, BBL, CPL** and other major cricket franchises.

05 DAO Governance

Transitioning to a **DAO structure** for entirely decentralized protocol upgrades, slashing parameters, and rulesets.

06 Integrity Partnerships

Official strategic partnerships and data-sharing agreements with cricket integrity bodies like the **ICC ACU**.

THANK YOU // Q&A

"Match integrity shouldn't be a *trust exercise*.

It should be a *math problem*."